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Revolutionize cold chain: an AI/ML driven approach to overcome capacity shortages

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ABSTRACT

This research investigates how Artificial Intelligence (AI) and Machine Learning (ML) forecasting methodologies can be leveraged for cold chain capacity planning, specifically utilising Prophet and Seasonal Autoregressive Integrated Moving Average parametrised through grid search. In collaboration with Americold, the world's second-largest refrigerated logistic service provider, the study explores the challenges and opportunities in applying AI/ML techniques to complex operations covering 385 customers and a capacity of 73,296 pallet positions. We train and test several AI/ML and traditional statistical models using extensive data for every customer over 3.5 years. Based on the results, MAPE of 5.28% was achieved on the whole site level, and SARIMA outperformed ML models in most cases. Next, we show that developing and applying a Customer Segmentation Matrix has enabled more accurate forecasting and planning across various customer segments, addressing the issue of forecasting inaccuracies. This approach effectively improves forecasting inaccuracies, underscoring the significance of tailoring AI/ML models for demand forecasting within the cold-chain industry. Ultimately, this research presents an AI-driven approach that transcends mere forecasting, offering a practical pathway to manage capacity in light of the constraints.

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1. Introduction

A 'cold chain' is a logistical process that involves the transportation and storage of temperature-sensitive products within a specific temperature range to maintain their quality and safety. It includes a series of temperature-controlled processes from raw material acquisition to the delivery of products to end consumers (Khan and Ali 2021). Also, the ever-increasing demand for fresh agricultural products, pharmaceuticals, and biotechnological goods requires efficient and reliable cold chains (Yu and Xiao 2021). The deterioration of fresh food can lead to substantial economic losses and food safety issues (Min Wang and Zhao 2021). Cases of mishandling have resulted in significant product losses, like temperature-sensitive food items being discarded due to inadequate storage conditions (Nerbovig 2017; Tsang et al. 2018). The significance of the cold chain becomes even more apparent when preservation during long-distance transportation results in approximately 40% value loss in fruit and vegetable (Negi and Anand 2015). Such losses not only impact the economic viability of businesses but also highlight the critical need for effective cold chains in preserving product quality and

minimising waste (Yu and Xiao 2021). Hence, to address these challenges, cold chains have emerged as a solution, employing temperature-controlled activities to slow down the deterioration of perishable items, attract safety-conscious consumers, and ensure food safety. Despite the growing body of research on AI and ML applications in supply chain management, there remains a significant gap in their use for cold chain capacity planning. Previous studies have primarily focussed on logistical models, such as truck transportation and vehicle routing, or on the adoption of technologies like blockchain and IoT for monitoring and tracking purposes (Cai et al. 2013; Khodaei, Kayvanfar, and Haji 2022; Li, Lee, and Gharehgozli 2023). However, there is a noticeable lack of research specifically addressing the application of AI/ML forecasting techniques to optimise capacity planning within cold chains (Behzadi et al. 2018). This gap is critical because traditional forecasting methods often fail to capture the complexities of cold chain logistics, such as the varying temperature requirements and the perishability of goods (Soto-Silva et al. 2016). By developing and validating an AI/ML forecasting framework tailored for cold chains, our study aims to bridge this gap and provide a

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more accurate and efficient approach to capacity planning in this sector.

Capacity planning in cold chains is a complex task for several reasons. First, the products of cold chains typically possess shorter shelf lives and are highly susceptible to environmental factors such as temperature, humidity, and lighting intensity (Gormley, Brennan, and Butler 2000). Maintaining precise environmental conditions becomes paramount throughout the cold chain, requiring advanced refrigeration and dehumidification systems (Tsang et al. 2018). The real complexity arises from the broad range of ambient temperatures required, spanning from -25°C to $+10^{\circ}\text{C}$, depending on the product type (Lana, Tijskens, and Van Kooten 2005; Soyer et al. 2010). This means different types of food products have varying temperature requirements. Perishable items like dairy, seafood, and fresh produce require different storage conditions, making it challenging to maintain consistent temperatures across the cold chain. Second, food safety regulations mandate strict temperature controls (Weng et al. 2015). Non-compliance can lead to legal consequences, recalls, and damage to a company's reputation. Third, most items within the cold chain typically consist of fresh agricultural or perishable products, which inherently come with extended supply lead times that are difficult to adjust due to environmental factors (Behzadi et al. 2018). Due to the vast distance between the production base and the target market in cold chains, the lead time is not only long but also usually quite unstable (Cai et al. 2013). Also, managing activities such as harvesting, post-harvest processes, packing, processing, storage, and transportation can be highly challenging due to sudden surges in supply. This indicates that implementing a cold chain involves a long lead time and significant capital investment in refrigeration and preservation equipment (Behzadi et al. 2018; Min Wang and Zhao 2021). This gets more challenging since seasonal demand fluctuations and changes in consumer preferences can lead to uneven demand patterns, complicating inventory management and cold chain logistics (Prentice and McLachlin 2008).

Above all challenges, the profit margins in this domain tend to be relatively narrow which highlights the necessity for effective capacity management practices and the utilisation of contemporary decision-making technological solutions (Soto-Silva et al. 2016). As a result, many firms, like Driscoll's, outsource cold chain services to logistic service providers (LSPs) to reduce operational costs and allow agri-product suppliers and retailers to focus on their core business activities (Mariappan et al. 2023; Yu and Xiao 2021). This means the role of cold chain logistics service providers (LSPs) cannot be overlooked in ensuring the seamless functioning of

cold chains. These specialised LSPs bring expertise and dedicated resources, playing a crucial role in the successful implementation and management of temperature-controlled activities, which are essential for maintaining the integrity and quality of perishable items throughout the cold chain.

Additionally, the significance of cold chains goes beyond cost savings, where the lack of cold chains results in substantial value and quality losses (Yu and Xiao 2021). Recall the case of Driscoll's and Americold; the investments made in refrigeration facilities are pivotal in maintaining the quality and freshness of their products throughout the entire supply chain (Min Wang and Zhao 2021). Americold, a prominent refrigerated Logistic Service Provider (LSP), invests significantly in procuring and maintaining such facilities (Min Wang and Zhao 2021). These investments are crucial for maintaining the freshness and integrity of perishable goods during their journey through the supply chain.

Given the distinct characteristics of cold chains, the task of capacity planning becomes a daunting task. On the one hand, the time-sensitive nature and perishability of products within cold chains require prompt transportation and storage to mitigate spoilage and maintain product quality (Gormley, Brennan, and Butler 2000). This means the insufficient capacity of cold chains could cause significant economic losses and even lead to legal ramifications, product recalls, and reputational damage (Nerbovig 2017; Tsang et al. 2018). Furthermore, in contrast to the relatively rapid capacity adjustments in conventional supply chains, the process of modifying capacity within cold chains is arduous and time-intensive. This stems from the extensive investments required for advanced refrigeration and dehumidification systems required for storing and transporting temperature-sensitive goods (e.g. Dai et al. 2020; Min Wang and Zhao 2021). On the other hand, excessive under-utilisation of capacity within cold chains is expensive, given the substantial investments necessary for these specialised systems.

In summary, recognising the significant consequences of over-utilisation and under-utilisation in cold chains, accurate demand forecasting is vital in optimising operations and capacity planning within these chains. Moreover, accurate demand forecasting ensures the timely transportation and storage of perishable and time-sensitive items in cold chains, thereby preventing wastage and upholding product quality. In other words, demand forecasting for highly constrained operations requires advanced techniques. This is because traditional forecasting methods, often relying on regression models adjusted with historical data, frequently deliver less accurate forecasts than newer approaches like AI and ML. AI/ML

models excel in identifying complex non-linear patterns and exploring relationships with seemingly unrelated external data sources, resulting in improved decision-making across various sectors (Hastie et al. 2009). Adopting these advanced algorithms has successfully enhanced decision quality in diverse managerial contexts (Brynjolfsson and McAfee 2014; LeCun, Bengio, and Hinton 2015). Hence, state-of-the-art Artificial Intelligence (AI) and Machine Learning (ML) forecasting methods can effectively address the unique complexities of the cold chain. Despite the advancements in AI/ML applications in supply chain management, there remains a significant gap in their application to cold chain capacity planning. Existing studies often overlook the unique complexities of temperature-sensitive logistics, such as varying temperature requirements and strict regulatory controls. By critically analysing these limitations, our research proposes a novel AI/ML framework and tests its effectiveness in a real-world setting. By applying AI/ML techniques, the primary objective is to enhance demand forecasting accuracy, optimise capacity planning, and improve overall cold chain operations.

Hence, we formulate our research question as follows:

- **Research Question:** *How can AI/ML forecasting methodologies be leveraged for cold chain capacity planning?*

To address the research question, we develop a novel forecasting framework designed for capacity planning in cold chains, which is necessary to ensure the quality of temperature-sensitive food products. The foundation of our study lies in a comprehensive review of existing literature on food cold chain management, demand forecasting methods, and qualitative interviews with key stakeholders within a real case study.

In our forecasting approach, we utilise two distinct modelling techniques: Seasonal Autoregressive Integrated Moving Average (SARIMA) and Facebook Prophet (FP). We selected SARIMA due to its well-established power and widespread application among both researchers and practitioners. SARIMA models offer an interpretability advantage and are based on established statistical principles. Their capability to handle time series data characterised by clear seasonal patterns makes them particularly suitable for the cold chain context. Conversely, AI/ML models, such as Facebook Prophet (FP), excel in detecting intricate non-linear patterns and uncovering relationships with seemingly unrelated external data sources. This proficiency results in enhanced decision-making across various sectors (Hastie et al. 2009). To measure the accuracy of our models, we utilise the Mean Absolute Percentage Error (MAPE)

performance metric. The insights of our forecasting models are invaluable, enabling us to propose the creation of additional freezer capacity at the selected site of the case study. Additionally, we identify underutilised space in the cooler segments that could potentially be utilised to increase freezer capacity, thereby optimising resource utilisation.

As a real case study, we collaborate with Americold, the world's second-largest refrigerated logistics service provider (LSP), the 2021 IARW Global Top 25¹ list by the Global Cold Chain Alliance. Americold specialises in the ownership, operation, acquisition, and development of temperature-controlled warehouses, managing a vast network of 250 facilities across North America, Europe, Asia-Pacific, and South America, with an impressive overall capacity of approximately 1.5 billion cubic feet and a revenue of \$2.7 billion, as reported in the Form 10-K of Americold Realty Trust in 2021². By selecting a specific warehousing site as our minimum viable product, we extract extensive inventory data for every customer over a 3.5-year period. Leveraging this data, we propose segmentation criteria based on customer *inventory size* and *variability* (operationalized by the Coefficient of Variation for customer demand), enabling us to develop accurate forecasting models tailored to each segment.

Overall, our findings highlight the significance of AI/ML techniques in cold chains to manage temperature-sensitive food supply chains effectively. In line with the core idea that there is not a single universal algorithm or model that excels above all others (commonly referred to as the 'no free lunch' principle), the results of this study also demonstrate that certain algorithms perform better than others based on the characteristics of the SKUs and customer segmentation. The managerial implications provided by the framework provide valuable guidance that can be beneficial for making decisions about selecting forecasting models and planning for capacity.

The subsequent sections of this paper are structured as follows. We review the related literature concerning cold chains and the integration of novel technologies like AI/ML models within this domain. Following this, we elaborate on the methodology employed and detail the procedure for data collection. Finally, we end this paper by proposing the framework and discussing the results.

2. Literature review

In this section, we examine the relevant literature. First, we review the papers related to cold chains and emphasise the attributes that differentiate cold chains from other supply chains. Next, we discuss the adoption of technologies in cold chains and dynamic capacity planning. Following that, we analyse studies closely related papers

to our research that employ AI/ML models in the context of cold chain or fresh food supply chains. The review of these sections will help us to pinpoint the research gap and position the current study.

2.1. Cold chains

A cold chain refers to a systematically engineered process in which refrigerated or temperature-sensitive products are stored, transported, distributed, and sold in controlled low-temperature environments to ensure product quality and safety (Cang and Wang 2021; Meng et al. 2022; Mohsin and Yellampalli 2017). In the past decade, there has been significant attention to cold chains due to their expanding range of applications (Zheng et al. 2021). Cold chains have unique characteristics, one of which pertains to the types of items they handle. Typically, these supply chains deal with perishable and temperature-sensitive items like vegetables, fruits, fresh meat, fish, shrimp, and other raw foods (Meng et al. 2022). The quality of these products is a critical factor and a fundamental criterion for consumers' choices (Cang and Wang 2021). This means ensuring the preservation of freshness and product quality necessitates tighter delivery timeframes and controlled storage conditions, which lead to higher end-product quality and reduced losses due to spoilage (Dabbene, Gay, and Sacco 2008; Soto-Silva et al. 2016). Additionally, cold chains face the challenge of long and often unpredictable transportation times due to the considerable distance between the production source and the target market (Behzadi et al. 2018; Cai et al. 2013). This extended lead time makes fresh products susceptible to decay and deterioration. Moreover, end consumers are highly sensitive to both the retail price and the freshness of the products, leading to uncertain market demand (Cai et al. 2013; Cang and Wang 2021). Hence, these challenges underscore the critical role that cold chains play in preserving product quality and pricing, highlighting their complexity compared to other supply chains (Cai et al. 2013).

Numerous logistical models related to cold chains have been extensively examined in existing literature (e.g. Cai et al. 2013; Jedermann et al. 2014; Khodaei, Kayvanfar, and Haji 2022; Zheng et al. 2021). For instance, Jedermann et al. (2014) conducted an assessment of technical remedies and implementations to track the shelf-life of perishable commodities in cold-chain food logistics. Their study also focussed on devising subsequent chain operations to curtail product wastage throughout the food distribution process.

Cold chains exhibit unique attributes that can lead to elevated CO₂ emissions. For instance, these chains

involve the preservation and movement of products at low temperatures, typically at or below freezing levels (Saif and Elhedhli 2016). This prerequisite mandates the utilisation of refrigerated warehouses and vehicles, which consume significant energy for refrigeration, potentially amplifying the carbon footprint. Within this context, Saif and Elhedhli (2016) introduce a novel mathematical model for devising cold supply chains with environmental concerns in mind. This model accounts not only for CO₂ emissions arising from energy consumption but also includes the effects of refrigerant gas leakage. In the context of E-commerce, Rodríguez García et al. (2023) presents a framework for organising operational costs in retail and warehouse e-fulfillment strategies, enabling brick-and-mortar grocery retailers to evaluate their advantages and disadvantages. Applying the Time-Driven Activity-Based Costing (TDABC) methodology, the framework is informed by insights from two prominent European supermarkets employing these strategies.

Studies have highlighted that product perishability in cold chains plays a critical role in shaping logistics operations and the design of distribution networks (Dolgui et al. 2018; Hasani, Zegordi, and Nikbakhsh 2012; Van Kampen and Van Donk 2014; Viet, Behdani, and Bloemhof 2020). This means the inherent complexity of managing the cold chain for fresh fruits surpasses that of other supply chains (Soto-Silva et al. 2016). These complexities emphasise the importance of efficient management practices and the integration of contemporary decision-making tools (Ahumada and Rene Villalobos 2009; Akkerman, Farahani, and Grunow 2010). In the context of food and grocery logistics, Rodríguez García et al. (2022) propose the building blocks for the value proposition and logistics strategy of grocery pure players. The study demonstrates that including fresh and frozen products in a retailer's portfolio significantly impacts the fulfillment process. Specifically, the authors highlight the effects on dispatch time slots, vehicle types, delivery time windows, and the appropriate packing methods.

In a comprehensive literature, Soto-Silva et al. (2016) show that a majority of the relevant research related to the cold-chain management of fresh fruits primarily focussed on the areas such as truck transportation, vehicle routing, production allocation, and the planning and distribution network. Hence, there is a lack of studies focussing on applying AI/ML forecasting techniques for capacity planning in cold chains (Behzadi et al. 2018). This is the first study that not only proposes a new framework but also tests the validity of the framework in a real case study. The proposed framework can help to enhance the efficiency of cold chain operations,

enabling accurate capacity allocation and ensuring on-time deliveries, which is particularly vital in industries like food and pharmaceuticals, where product quality and safety are important.

2.2. Technologies in cold chains

Some studies have investigated the adoption of new and disruptive technologies in cold chains with a particular focus on fresh and agri-food items. For instance, Wu, Fan, and Cao (2023) examines the approaches to integrating blockchain technology into the cold chain associated with the supply of fresh products. They demonstrate that embracing blockchain technology might not always be the most advantageous choice. This is because the outcome is tied to factors such as consumer acceptance of products without blockchain technology, the rate of fresh product deterioration, and the distribution of traceability costs among supply chain participants upon adopting blockchain technology. Li, Lee, and Gharehgozli (2023) examine and evaluate the primary blockchain platforms utilised within cold chains that are related to food supply networks. Through an exhaustive synthesis analysis, they explore the benefits and challenges linked with blockchain technology. The study underscores that blockchain improves visibility at every phase of the cold chain, increases transaction transparency, food safety, and quality, and simultaneously alleviates concerns regarding food fraud and wastage. Additionally, blockchain serves as a digital solution to reducing operational costs and enhancing efficiency within food supply chains (Li, Lee, and Gharehgozli 2023).

Defraeye et al. (2021) examine how Digital Twins (DT) can be useful in managing cold chains. They emphasise that DT provides valuable information to exporters, retailers, and consumers. This information includes specifics on the time left before a shipment's products expire, which can guide decisions about logistics and marketing. These technologies also help in spotting and predicting possible problems in supply chains that might lead to reduced food quality and waste.

Anticipatory shipping (AS) tools have been used in the context of cold chains (Viet, Behdani, and Bloemhof 2020). AS uses historical data to predict future orders and deliver products to nearby distribution centres before customers order, balancing customer demands and cost efficiency (Lee 2017). Viet, Behdani, and Bloemhof (2020) apply AS for cold chains related to agro-food items, addressing product perishability and imbalance issues. By integrating product quality data into analysis and optimising production and transportation processes, the research shows anticipatory shipping can

improve delivery service by up to 35.3% and cut costs by up to 9.3%, as validated through simulations using a Dutch floriculture supplier's data.

Internet of Things (IoT) devices offer valuable assistance in data collection by continuously and accurately gathering real-time information from various sources (Ko et al. 2015). Considering the unique characteristics of cold chains, IoT plays a significant role in simplifying processes such as gathering data, supervising product quality, managing logistics, and enhancing payment efficiency (Ruan and Shi 2016). The IoT devices enable real-time monitoring of temperature, humidity, and other critical parameters, ensuring the preservation of perishable goods and enhancing overall supply chain visibility and efficiency. In this context, Tsang et al. (2021) present an IoT system for planning deliveries with multiple temperature requirements (IoT-MTDPS). This system incorporates a two-stage multi-objective genetic algorithm optimiser (2PMGAO). By utilising IoT-MTDPS, the capacity to manage e-commerce orders is improved while ensuring customer contentment at a specified standard. Hence, established and continually expanding technologies such as the Internet of Things (IoT), cloud computing, and Wireless Sensor and Actuator Networks (WSANs) have offered unique opportunities to remotely control and monitor various regulated parameters in cold chain with minimal human intervention (Mohsin and Yellampalli 2017). Feng et al. (2019) employed a WSN network to dynamically monitor various quality parameters such as temperature, humidity, O₂, and CO₂ to enhance the safety of frozen shellfish in the cold chain supply chain. Lastly, Torres-Sánchez et al. (2020) introduced a real-time monitoring system that employed multiple non-linear regression techniques to predict the shelf life of fruits and vegetables based on factors such as temperature, relative humidity, and gas concentration.

Kinetic analysis, using past data and mathematical models or machine learning tools, is a helpful tool in numerous fields and industries, helping make future plans and strategies and generally forecasting. In the context of cold chains, Roduit et al. (2019) employed an advanced kinetic analysis to forecast the shelf life of cold chain items through the monitoring of time-temperature data using a data-logger. Yujun Wang et al. (2015) introduced a monitoring and decision system that leveraged a wireless sensor network (WSN) and an ontology-based knowledge representation approach to evaluate the quality of cold chain products. The proposed approach interprets data like temperature, humidity, and location directly obtained from sensors within a refrigerated truck to diagnose product status and trigger appropriate alerts.

2.3. Dynamic capacity planning

As noted in the literature, capacity planning plays an important role in improving business performance in industries with high capital investment costs (Al-Shobaki and Mohsen 2008; Geng and Jiang 2009; Uzsoy, Fowler, and Mönch 2018). Similarly, the huge investments required for advanced refrigeration and dehumidification systems in the cold chains industry (Behzadi et al. 2018; Dai et al. 2020), make capacity planning very relevant. Precise demand prediction is crucial for making decisions like capacity planning, labour scheduling, and production planning (Uzsoy, Fowler, and Mönch 2018).

Although studies have highlighted that forecasting is a crucial component of any planning process, a comprehensive capacity planning process requires more than just forecasting (Grimson and Pyke 2007; Noroozi and Wikner 2017; Thomé et al. 2012). This is due to several reasons. First, forecasts are not always entirely accurate, particularly given the frequent occurrence of black swan events (Taleb 2005) that are drastically affecting supply chains currently (Dolgui and Ivanov 2021; Dolgui, Ivanov, and Sokolov 2018; Ivanov, Dolgui, and Sokolov 2019). Second, even without these events, forecasts inherently lack the data points necessary for precise projections. For example, traditional forecasting methods often struggle to capture and handle the random fluctuations in demand effectively (Huang, Chang, and Chou 2008). The inaccurate forecasts, in turn, impact the accuracy and efficiency of subsequent production scheduling and capacity planning (Huang, Chang, and Chou 2008). Third, even if a company possesses every data point at the present moment, it would be unrealistic to assume that each data point remains unchanged as inputs into a future forecast. In a similar line, the famous phrase ‘All models are wrong!’ by Box (1976) is a common sentiment and has become even more prevalent with these unexpected events.

Therefore, the primary goal of this paper is to develop a dynamic and comprehensive capacity planning framework designed for cold chains. This process extends the demand forecasting process and includes the capacity to adjust and optimise resources, such as storage, customer segmentation, and labour planning. The provided framework considers various factors such as site-level forecasting, room-level forecasting, and customer segmentation. This comprehensive approach is aimed at achieving capacity planning that is both effective and flexible within the context of cold chain logistics.

2.4. AI/ML forecasting in cold chains

To the best of our understanding, a gap exists in utilising AI/ML models for the capacity planning of cold

chains. Previous research has predominantly focussed on using AI/ML tools in predicting time and monitoring temperature (Aung and Chang 2014; Awad, Ndiaye, and Osman 2020; Göransson, Jevinger, and Nilsson 2018). Mariappan et al. (2023) use AI/ML techniques to predict shipment times in cold chains. They analyse over 3 million real-world shipments from an e-pharmacy, creating a diverse ensemble of AI/ML models for improved accuracy. Their approach outperforms existing methods and proves effective for forecasting shipment times within the cold chain related to the e-pharmacy supply chain. He and Yin (2021) focus on cold chain logistics in the Chinese market and develop demand prediction models using AI/ML models. They use neural networks and grey prediction to study cold chain logistics demand. The authors show the neural network algorithm has slightly better accuracy, making it the preferred choice for demand forecasting in cold chain logistics management. Cannas et al. (2023) explore the utilisation of artificial intelligence (AI) in Operations and Supply Chain Management (OSCM) by analysing empirical data from 17 AI applications within six Italian companies. The study assesses how AI applications enhance various processes within OSCM and examines the advantages and challenges companies encounter during their adoption.

In the food supply chain, Priyadarshi et al. (2019) show that AI/ML tools, precisely the use of long short-term memory (LSTM) and support vector regression (SVR), for demand forecasting of certain vegetables, leads to improved outcomes. These enhancements are observed in terms of inventory turnover and days of stock coverage, which aid in averting stock shortages and reducing supply chain volatility. Although the results of the study cannot be generalised, they can still be used for forecasting various agricultural products at the retail stage, taking into account the distinct demand characteristics of each item. Haselbeck et al. (2022) compared nine modern AI/ML techniques with three traditional forecasting methods to predict horticultural sales. The results consistently favoured AI/ML models, especially XGBoost, which came out on top in 14 out of 15 comparisons. Also, the authors demonstrate that including additional external factors, such as weather and holiday information, as well as meta-features, will improve the model’s accuracy. In addition, they examined whether the algorithms could capture the sudden increase in demand of horticultural products during the SARS-CoV-2 pandemic in 2020, and XGBoost outperformed all other models.

To the best of our knowledge, there is an existing gap in leveraging AI/ML models for capacity planning in cold chain logistics. This research introduces a novel AI/ML framework and validates its efficacy through a practical case study. The framework is based on a multi-level

forecasting approach, which includes site-level, room-level, and customer segmentation. The results highlight that capacity planning in the cold chain is challenging; however, the framework can enable businesses to determine the right inventory levels, optimise resource allocation, ensure timely deliveries, and enhance cost efficiency. This is critical in time and temperature sensitive industries like food and pharmaceuticals, where product quality and safety are essential.

3. Methodology

This section presents the methodology behind the proposed cold chain capacity planning framework. We start by describing the dataset, the implementation of AI/ML for forecasting, the approach behind customer segmentation, and the eventual development of the capacity planning framework. As previously discussed, there is a noticeable gap of research that applies AI/ML techniques to the capacity planning of cold chains. This study aims to bridge this gap by collaborating closely with Americold. Our approach involves an initial step of segmenting customers into four distinct quadrants, which we achieve by utilising the coefficient of variation (CV) and analysing the inventory levels occupied by each customer. This segmentation strategy allows us to account for the diverse demand patterns exhibited by different customer segments. Subsequently, we apply a range of AI/ML models to each customer segment, tailoring our predictions to the unique requirements of each segment. Given the significant variations in demand patterns across different customer types, adopting a one-size-fits-all prediction model is not feasible, and thus, our approach ensures that we deploy the best-suited prediction model for each customer segment.

3.1. Data

To address the research question and test the proposed methodology, we collaborate with Americold, the world's second-largest refrigerated LSP. The company specialises in the ownership, operation, acquisition, and development of temperature-controlled warehouses, managing a vast network of 250 facilities across North America, Europe, Asia-Pacific, and South America, with an impressive overall capacity of approximately 1.5 billion cubic feet and a revenue of \$2.7 billion. The dataset under consideration covers only one site with a total capacity of 73,296 pallet positions. The site contains both cooling chambers, with a total capacity of 13,632 pallet positions, and freezing chambers, with a total capacity of 59,664 pallet positions. The dataset covers the period from 14 June 2019 to 19 February 2023 and includes 385 individual customers.

The comprehensive dataset from Americold, encompassing diverse customer profiles and detailed transactional data over 3.5 years, offers a robust framework for assessing the efficacy of AI/ML models in the context of complex, real-world cold chain logistics, making it highly suitable for the scope of our study.

3.2. Forecasting algorithms

Forecasting is central to many activities within an organisation. For instance, organisations across all sectors of industry must engage in capacity planning to efficiently allocate scarce resources and establish a mechanism to measure and track performance in relation to a predefined baseline (Taylor and Letham 2018). The algorithms under consideration include SARIMA and Prophet. In the subsequent experiments, we aim to juxtapose the performance of a state-of-the-art AI/ML forecasting algorithm, Prophet, with that of SARIMA, a traditional and well-established approach. This comparison elucidates the relative strengths and practical implications of advanced versus conventional methodologies within the cold chain capacity planning context.

3.2.1. Seasonal autoregressive integrated moving average

ARIMA (p, d, q) model is described by Equation (1). In this formulation, p corresponds to the order of the autoregressive component, d is the degree of differencing, and q is an order of a moving average component (Box et al. 2015).

$$y'(t) = c + \phi_1 y'(t-1) + \dots + \phi_p y'(t-p) + \theta_1 \epsilon(t-1) + \dots + \theta_q \epsilon(t-q) + \epsilon(t) \quad (1)$$

where $y'(t) = y(t) - y(t-1)$ is the differenced series, the autoregressive component of order p (AR(p)) is described by Equation (2), and the moving average component of order q (MA(q)) is described by Equation (3).

$$y(t) = c + \phi_1 y(t-1) + \dots + \phi_p y(t-p) + \epsilon(t) \quad (2)$$

$$y(t) = c + \epsilon(t) + \theta_1 \epsilon(t-1) + \dots + \theta_q \epsilon(t-q) \quad (3)$$

where $\epsilon(t)$ is a white noise (Hyndman and Athanasopoulos 2018). The ARIMA model can be represented in a more compact way using the backshift operator $By(t) = y(t-1)$. Using the backshift operator, the first-order difference can be transformed into $y'(t) = y(t) - y(t-1) = y(t) - By(t) = (1-B)y(t)$, therefore Equation (1) can be rewritten as Equation (4).

$$(1 - \phi_1 B - \dots - \phi_p B^p)(1 - B)^d y(t) = c + (1 + \theta_1 B + \dots + \theta_q B^q) \epsilon(t) \quad (4)$$

This formulation is critical to define the seasonal extension of ARIMA (SARIMA). SARIMA is formed by

adding additional seasonal terms to the ARIMA models, which can be represented as follows: SARIMA $(p, d, q)(P, D, Q)_m$, where m corresponds to the number of observations per year, for example, $m = 52$ in our study because we deal with data aggregated by week (Hyndman and Athanasopoulos 2018). The lower-case notation is for the non-seasonal parts of the model, and the uppercase notation is used for the seasonal parts ones. For example, SARIMA $(1, 1, 1)(1, 1, 1)_{52}$ corresponds to Equation (5).

$$\begin{aligned} (1 - \phi_1 B)(1 - \Phi_1 B^{52})(1 - B)(1 - B^{52})y(t) \\ = (1 + \theta_1 B)(1 + \Theta_1 B^{52})\epsilon(t) \end{aligned} \quad (5)$$

We selected the SARIMA model using an exhaustive grid search (Jiménez, Lázaro, and Dorronsoro 2008).

3.2.2. Facebook prophet

Prophet is a Bayesian AI/ML algorithm developed by Facebook's Core Data Science team. Prophet aims to decompose time-series data $y(t)$ into three main components: a trend component $g(t)$, a seasonality component $s(t)$, and a holiday effect component $h(t)$ (Taylor and Letham 2018). Our application doesn't use $h(t)$ due to the data aggregation and the fact that the variability in capacity utilisation during major holidays can be captured within $s(t)$. Therefore, the Facebook Prophet model is reduced to a decomposable time-series (Harvey and Peters 1990) of the form described by the Equation (6).

$$y(t) = g(t) + s(t) + \epsilon_t \quad (6)$$

where the error term, represented by ϵ_t , stands for any idiosyncratic changes not covered by the model. It is assumed within the model that ϵ_t follows a normal distribution.

This particular configuration bears a resemblance to a generalised additive model (Hastie and Tibshirani 1987), which constitutes a group of regression models where potentially nonlinear smoothers are applied to the regressors. In this context, only time is used as a regressor, but potentially several linear and nonlinear functions of time serve as components. The methodology of modelling seasonality as an additive component echoes the approach employed by exponential smoothing, as detailed by Gardner Jr (1985).

The trend is modelled as a piece-wise constant rate of growth (Equation (7)).

$$g(t) = (k + a(t)^T \delta)t + (m + a(t)^T \gamma) \quad (7)$$

where k is the growth rate, m is the offset parameter. Adjustments vector $a(t)$ with the corresponding rates of adjustments δ and γ are responsible for the trend changes and are defined below.

Trend changes are modelled by explicitly defining changepoints where the growth rate is allowed to change. Given S changepoints at times s_j , $j = 1, \dots, S$. A vector of rate adjustments can be defined using $\delta \in \mathbb{R}^S$, where δ_j is the change in rate that occurs at the moments of time s_j . The rate at t is then considered as the base rate k , plus all of the adjustments up to that point, namely $k + \sum_{j:t > s_j} \delta_j$. This procedure can be represented by defining a vector $a(t) \in 0, 1^S$ according to Equation (8) (Taylor and Letham 2018).

$$a_j(t) = \begin{cases} 1, & \text{if } t \geq s_j, \\ 0, & \text{otherwise.} \end{cases} \quad (8)$$

Based on that, the rate at time t can be calculated as $k + a(t)^T \delta$. It is important to note that When the rate k is adjusted, the offset parameter m must also be adjusted. The adjustment at changepoint j is made using the coefficient γ_j that can be calculated according to he Equation (9) (Taylor and Letham 2018).

$$\gamma_j = \left(s_j - m - \sum_{l < j} \gamma_l \right) \left(1 - \frac{k + \sum_{l < j} \delta_l}{k + \sum_{l \leq j} \delta_l} \right) \quad (9)$$

Facebook Prophet relies on the Fourier series (Harvey and Shepard 1993) to construct a seasonal component $s(t)$. Setting P as the regular period we expect the time series to have, the arbitrary smooth periodic effects can be approximated according to Equation (10):

$$s(t) = \sum_{n=1}^N \left(a_n \cos \frac{2\pi nt}{P} + b_n \sin \frac{2\pi nt}{P} \right) \quad (10)$$

Fitting $s(t)$ is conducted through estimating the vector of parameters $\beta = [a_1, b_1, \dots, a_N, b_N]$ (Taylor and Letham 2018). $P = 52$ in our case because we deal with yearly data aggregated by week.

3.2.3. Validation

In planning capacity for the cold storage company, single-point forecasts are of little utility, and it is essential to make forecasts over a certain horizon H .

Each of H estimates of the future data points is associated with some error. Therefore, it is essential to specify empirical metrics that may serve to measure forecasting accuracy, compare methods and track the performance of the forecasting model overtime. Additionally, such a metric helps to diagnose how error-prone the forecasting procedure is, which in its turn, allows capacity planning team to determine whether the forecast is trustworthy at all and to which extent. Such a metric can be formally defined as the empirical accuracy of a forecast of

$h \in (0, H]$ periods ahead of time T (Equation (11)).

$$\phi(T, h) = d(\hat{y}(T + h | T), y(T + h)) \quad (11)$$

where $\hat{y}(T + h | T)$ represents a forecast for h steps ahead of T made using historical observations up to time T and $d(\hat{y}(T + h | T), y(T + h))$ is a distance metric used to measure the forecasting accuracy (Taylor and Letham 2018).

We set the forecast horizon to 26 weeks in the following analysis ($h = 26$). As highlighted by De Gooijer and Hyndman (2006), the choice of a distance metric must be problem-specific. De Gooijer and Hyndman (2006) and Hyndman and Koehler (2006) provide a detailed overview of error metrics suitable for forecasting. In our study, we decided to focus on mean absolute percentage error (MAPE) presented in Equation (12).

$$MAPE = \frac{100\%}{T} \sum_{t=1}^T \frac{|y_t - \hat{y}_t|}{y_t} \quad (12)$$

In the field of AI/ML forecasting, mainly when applied to forecasting tasks within supply chain and operations management, MAPE stands out as the most commonly adopted metrics (Chou et al. 2023; Chuang, Chou, and Oliva 2021; Zhu et al. 2021). MAPE, favoured for its interpretability in AI/ML models, does come with certain caveats. Notably, when actual values (y_t) are zero or approaching zero, MAPE can become infinite or severely skewed, a concern highlighted by Gardner (1990). Additionally, MAPE's inherent structure means it penalises positive errors more than negative ones, introducing an asymmetry, as pointed out by Makridakis (1993). However, in AI/ML applications where all data points are positive and significantly distant from zero, these limitations become less consequential. As Hyndman and Koehler (2006) concluded, under such conditions, MAPE remains a reliable metric.

3.2.4. Prediction intervals from bootstrapped residuals and capacity breaks estimation

As highlighted by Box (1976) 'All models are wrong!'. That is why the forecasting errors and the corresponding prediction intervals are central for capacity planning under uncertainty. A prediction interval for forecast $\hat{y}_{T+h|T}$ can be written as $\hat{y}_{T+h|T} \pm c\hat{\sigma}_h$, where $\hat{\sigma}_h$ is an estimate of the standard deviation of the h -step forecast distribution and c is the multiplier to control the width of the interval (Hyndman and Athanasopoulos 2018).

In order to estimate $\hat{\sigma}_h$ without making additional assumptions on the models and the distribution of residuals, we build prediction intervals from bootstrapped residuals. If forecast error is defined as $e_t = y_t - \hat{y}_{t|t+1}$,

we can solve it for y_t and rewrite as follows $y_t = \hat{y}_{t|t+1} + e_t$. In this setting, we can simulate the next observation of a time series as follows $y_{T+1} = \hat{y}_{T+1|T} + e_{T+1}$, where $\hat{y}_{T+1|T}$ is the one-step forecast and e_{T+1} is the future error (Thombs and Schucany 1990). Assuming future errors will be close to past errors, e_{T+1} can be sampled from the past residuals. Adding the new simulated observation to our data set, we can repeat the process to obtain $y_{T+h} = \hat{y}_{T+h|T} + e_{T+h}$ and subsequently $\hat{\sigma}_h$. As a result, if y_{T+h} is viewed as a random variable $y_{T+h} = X \sim N(\hat{y}_{T+h|T}, \hat{\sigma}_h)$, its cumulative distribution function (CDF) corresponds to the probability that the site's total capacity or target capacity x will be sufficient $F_X = P(X \leq x)$ (Rohatgi and Saleh 2015). Based on that $1 - F_X = P(X \geq x)$ corresponds to the probability that the site will break capacity. Figure 1 illustrates the intuition behind this estimation. In a practical setting, the normality of residuals has to be tested using the Anderson-Darling (Anderson and Darling 1954) Normality test (or alternative test) prior to further probabilistic reasoning.

At this stage, it is essential to once again explain the difference between total capacity and target capacity. The total capacity is the theoretical maximum capacity of the site restricted by the total cubic volume available across all the chambers. However, if the total capacity is reached, the picking and handling becomes substantially harder to perform, which slows the operation and reduces the overall efficiency. That is why in practical settings, the site uses the notion of target capacity (91% of the theoretical maximum for Americold), which is the optimal upper bound that allows the site to perform without jeopardising its efficiency.

3.3. Customer segmentation

We developed the Customer Segmentation Matrix inspired by the Growth Share Matrix (also known as BCG matrix). The Growth Share Matrix is a two-by-two matrix created by the Boston Consulting Group in the early 1970s as a tool to help companies with the task of portfolio management (Reeves and Moose 2023). It allows companies to manage different products in their portfolio based on two dimensions: market growth rate and market share (Hax and Majluf 1983; Seeger 1984).

In our model, we replaced these factors with inventory size (which is a primary revenue driver) and variability as our two main criteria for customer segmentation (Figure 2). To determine inventory size, we computed the average inventory during the analysis year. For variability, we utilised the Coefficient of Variation (CV) for outbound demand. The CV is calculated by dividing the standard deviation by the mean, indicating the dispersion

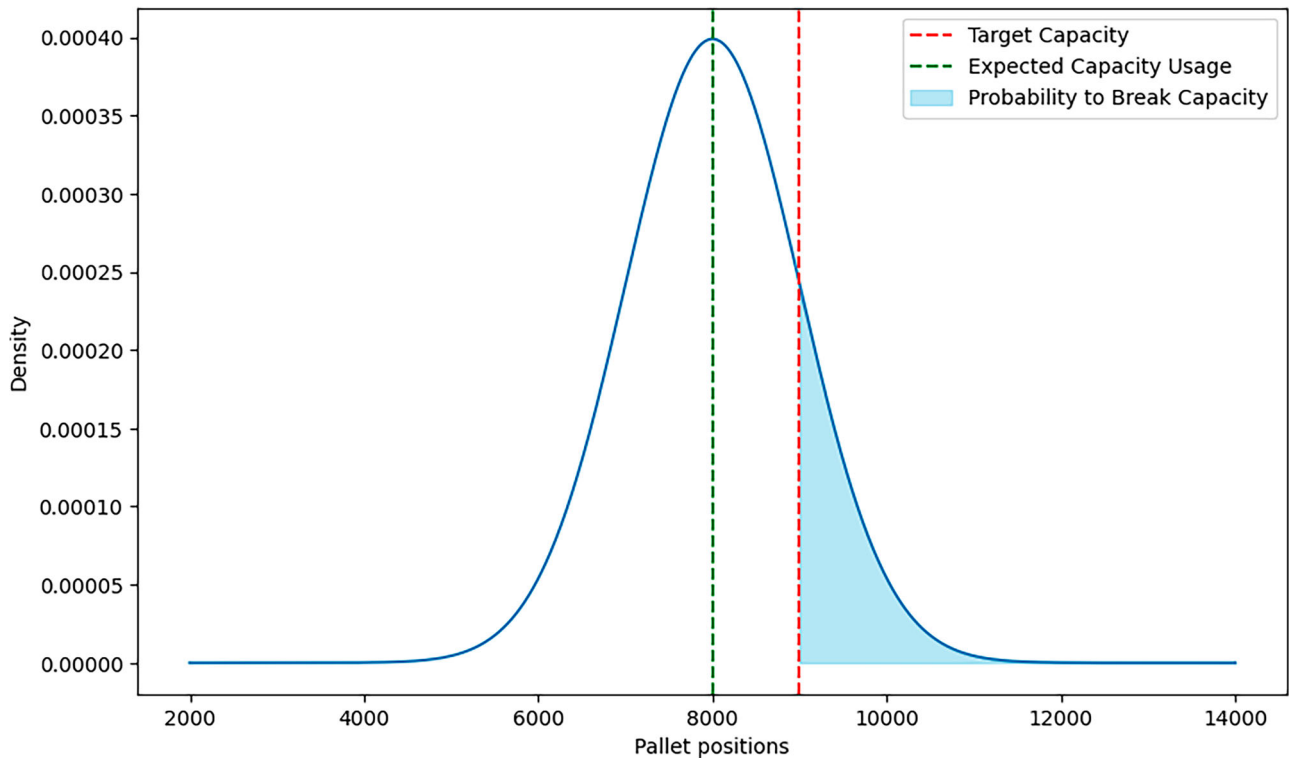


Figure 1. Probabilistic view of the capacity. The shaded area corresponds to the probability that the site will break capacity x ($1 - F_X = P(X \geq x)$).

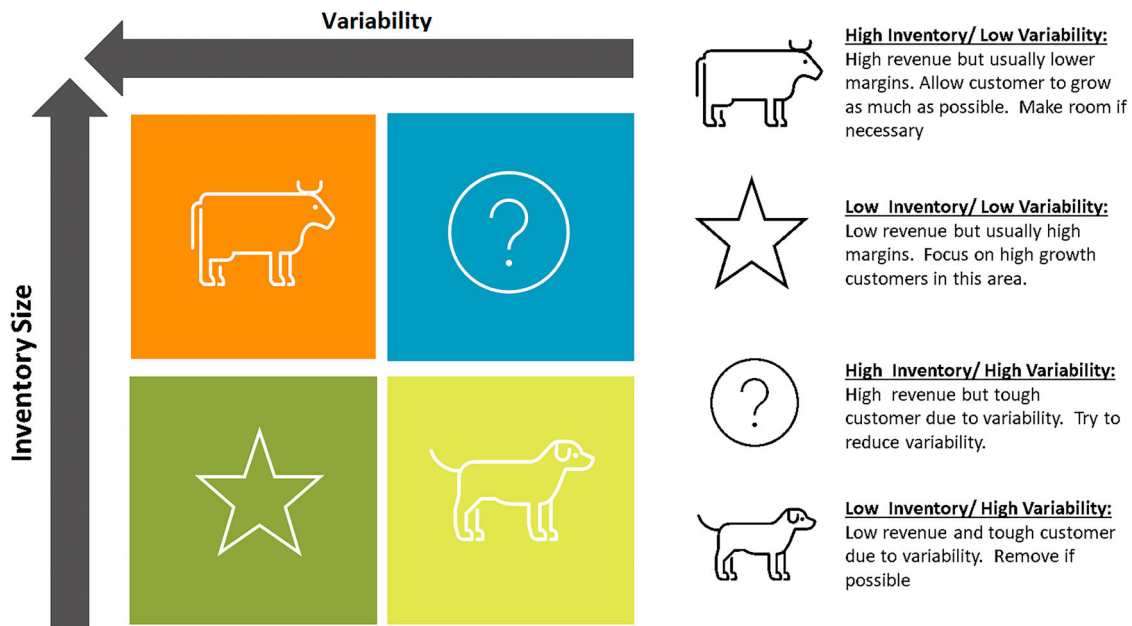


Figure 2. Customer Segmentation Matrix inspired by the growth-share matrix.

around the mean. The lower the dispersion, the more stable and potentially predictable the volume is on any given day. We selected outbound demand for this metric as it is the most labour-intensive activity in a distribution centre.

CV is a widely recognised metric in academia (Chou et al. 2023; Chuang, Chou, and Oliva 2021; Zhu

et al. 2021) and industry (Hopp and Spearman 2011) for measuring volatility, especially in demand forecasting and inventory management. Its adoption stems from its ability to provide a normalised measure of dispersion, allowing for a more consistent comparison across different scales and units. This fact makes CV particularly

valuable in assessing the relative variability of demand patterns, aiding businesses in making more informed inventory decisions.

These two dimensions are then used to classify products or business units into four categories:

- (1) High inventory / Low variability (Cash Cows). High revenue but usually lower margins. The proposed business strategy is to allow customers to grow as much as possible and reserve additional storage capacity if necessary.
- (2) Low inventory / Low variability (Stars). Low revenue but usually high margins. The proposed business strategy is to focus on the customers with the potential to grow and become a 'Cash Cow'.
- (3) High inventory / High variability (Question Marks). These customers generate high revenue but are hard to deal with due to the high variability and, therefore, unpredictability. The proposed business strategy is to reduce variability through commitment agreements and relationship management.
- (4) Low inventory / High variability (Dogs). These customers don't contribute much to the revenue and are hard to deal with due to high variability. The proposed business strategy is to charge them above-average tariffs and potentially remove them to create space for more promising customers.

3.4. Capacity planning framework

The proposed capacity planning framework is designed in accordance with the principles of CRISP-DM (Cross-Industry Standard Process for Data Mining), a well-established, structured process model for planning and executing data mining projects (Shearer 2000). The elements of the capacity planning framework for Cold Chain can be connected with various stages of the CRISP-DM model as follows:

- (1) Business Understanding: This phase delves deep into understanding the objectives and requirements of the project from a business perspective. In the context of cold chain logistics, this is crucial due to the sensitivity of products and the high costs associated with temperature deviations. Within the capacity planning framework, this phase aligns with the literature review stage, which comprehends the business problem of the cold chain industry's intricacies. Feedback from industry experts further refines the understanding of business requirements and objectives.
- (2) Cold chain logistics is data-intensive, with variables like temperature, humidity, and transit times playing

pivotal roles. This phase involves the collection and initial exploration of such data. Within the capacity planning framework, it corresponds to the data analysis stage (segmentation and aggregation) and qualitative interviews. These processes ensure a comprehensive understanding of data, identifying data quality issues and exploring data properties and relationships.

- (3) Data Preparation: Given the critical nature of cold chain logistics, ensuring data accuracy is paramount. This phase, in the CRISP-DM model, involves cleaning, formatting, and integrating data. Within the capacity planning framework, this aligns with the data validation phase, ensuring the data is accurate and primed for further processing.
- (4) Modelling: This stage is vital for predicting potential disruptions or inefficiencies in the cold chain. It encompasses the selection and application of various models to the prepared data. Within the capacity planning framework, this phase is represented by developing and applying forecasting models tailored to cold chain nuances.
- (5) Evaluation: Ensuring that models are robust and can handle the complexities of cold chain logistics is crucial. This stage in the CRISP-DM process involves assessing the models against the business objectives and requirements identified earlier. Within the capacity planning framework, this aligns with the steps involving exogenous factors and feedback from cold chain experts.
- (6) Deployment: The final phase is where the rubber meets the road. In the context of cold chains, accurate capacity planning can mean the difference between product integrity and spoilage. Within the capacity planning framework, this phase is represented by refining forecasting models to produce actionable business insights. This step applies the model's findings to make informed decisions, ensuring optimal cold chain operations.

Connecting the elements of the capacity planning framework with the stages of the CRISP-DM model ensures a structured and effective approach to data mining projects.

4. Results

The following section illustrates the application of the proposed framework for capacity planning based on the real data provided by Americold. The section demonstrates how customer segmentation and AI/ML forecasts could be streamlined to facilitate critical decision-making regarding capacity planning.

Table 1. Accuracy of best-performing forecasting algorithms at the level of the whole site, freezer, cooler, and different customer segments.

Level of aggregation	Best algorithm	Best MAPE *	2nd Best algorithm	2nd Best MAPE*	Delta**	MA***
Whole site	SARIMA m(12) (1,0,1) (1,1,1)	5.28% (2.09)	Prophet	5.87% (3.11)	0.59%	6.5%
Cooler	SARIMA m(52) (0,1,0)(1,0,0)	11.73% (4.21)	SARIMA m(12) (1,0,0)(1,1,0)	11.92% (5.92)	0.19%	13.5%
Freezer	Prophet	6.11% (1.87)	SARIMA m(52) (0,1,0)(1,0,0)	11.73% (4.21)	5.62%	8.0%
Key Customer 1	SARIMA m(12) (2,1,1)(0,1,1)	13.17% (6.29)	SARIMA m(52) (0,1,1)(0,1,1)	13.91% (6.15)	0.74%	15.5%
Key Customer 2	SARIMA m(12) (1,0,2)(0,1,1)	12.84% (5.20)	SARIMA m(52) (1,0,1)(0,1,0)	15.11% (3.26)	2.27%	14.5%
Key Customer 3	SARIMA m(52) (1,0,1)(0,1,1)	14.55% (3.99)	SARIMA m(12) (2,0,0)(1,1,0)	14.58% (3.42)	0.03%	16.0%
Low_Inv-High_CV_COOLER	SARIMA m(52) (0,1,0)(1,0,0)	20.62% (4.90)	SARIMA m(12) (2,0,2)(1,0,1)	21.69% (5.56)	1.07%	25.0%
Low_Inv-High_CV_FREEZER	SARIMA m(52) (1,0,1)(1,1,1)	9.90% (2.29)	SARIMA m(12) (2,0,0)(0,1,1)	12.38% (3.65)	2.48%	15.0%
Low_Inv-Low_CV_COOLER	SARIMA m(12) (1,0,1)(1,1,0)	17.79% (11.17)	SARIMA m(52) (0,1,0)(1,0,0)	17.80% (7.87)	0.01%	20.0%
Low_Inv-Low_CV_FREEZER	SARIMA m(52) (1,0,1)(1,0,0)	16.86% (6.32)	SARIMA m(12) (1,0,0)(0,1,1)	19.66% (2.38)	2.8%	18.5%
MVC Program_COOLER	SARIMA m(12) (1,0,1)(0,1,1)	11.86% (8.93)	Prophet	14.13% (10.57)	2.27%	14.0%
MVC Program_FREEZER	Prophet	7.65% (2.00)	SARIMA m(52) (0,1,0)(1,0,0)	9.01% (5.15)	1.36%	10.0%

* Parentheses contain standard deviation of MAPE

** Delta represents the absolute difference in MAPE between 1st and 2nd best forecasting algorithms

*** MA corresponds to the 4-weeks rolling average used as a benchmark

4.1. Forecasting-based capacity planning

Since historical data is available up to February 19, 2023, and the forecast horizon is set to 26 weeks in the following analysis ($h = 26$), the forecast is made until August 20, 2023, for all the temperature controls and customer segments. This 26-week horizon has been chosen in line with the capacity planning context. In industries where inventory and storage play a pivotal role, having a medium-term outlook is crucial to ensure optimal utilisation of resources and to anticipate any potential challenges or changes in demand (Gambaro et al. 2023). Furthermore, referring to the current managerial practices of Americold, a 26-week forecast aligns well with their biannual review cycles, allowing them to make informed decisions based on the most recent data and trends. We first identify the most accurate AI/ML model based on the time-series cross-validation. Table 1 summarises the results for the whole site as well as for the freezer and cooler.

The site has a total capacity of 73,296 pallet positions and a target capacity of 66,846 pallet positions. At this stage, it is essential to once again explain the difference between total capacity and target capacity. The total capacity is the theoretical maximum capacity of the site restricted by the total cubic volume available across all the chambers. However, if the total capacity is reached, the picking and handling becomes substantially harder to perform, which slows the operation and reduces the overall efficiency. That is why in practical settings, the site uses the notion of target capacity (91% of the theoretical maximum), which is the optimal upper bound that allows the site to perform without jeopardising its efficiency.

The algorithm forecasts that on August 20, 2023, there should be customer demand for 61598 pallet positions ($\hat{y}_{T+26|T}$). Given the $\hat{\sigma}_{26}$ of 7162 pallet positions, calculated using bootstrapped residuals, the probability of breaking the target capacity is 23.2%, and the probability of breaking the total capacity is 5.1%. Taking these

numbers into consideration, the capacity planning team has to seriously consider expanding the site or building a new one (Figure 3). The practical limitation of reaching maximum capacity must be underscored, owing to the management of customers via negotiations and pricing strategies. Despite this constraint, the projected customer demand, coupled with a 23.2% probability of breaking the target capacity and a 5.1% chance of surpassing the total capacity, creates a critical situation. It could result in an inability to accommodate new customers or necessitate charging existing customers beyond standard rates to encourage their relocation to alternative sites. Such a scenario emphasises the urgent need for the capacity planning team to contemplate either the expansion of the current facility or the development of a new one.

The freezer chambers, in total, occupy enough space to allocate 59,664 pallet positions, with the target capacity at 53,630 pallet positions. It is essential to point out that both the total and target capacity of a freezer increased on November 12, 2022, because the site management decided to convert one of the cooling chambers into a freezing one (Figure 4).

The algorithm forecasts that on August 20, 2023, there should be customer demand for freezing chambers equivalent to 53,590 pallet positions. Given the sigma of 5320 pallet positions, calculated using bootstrapped residuals, the probability of breaking the target capacity is 49.7%, and the probability of breaking the total capacity is 12.7%. These projections underscore the significant risk of breaking the target capacity. In the short term, the capacity planning team must contemplate converting one of the cooling chambers to freezing as an immediate remedial measure. However, given the statistical trends, it becomes evident that a long-term solution will necessitate not merely a modification of the current infrastructure but an expansion of the existing site or the construction of an entirely new facility.

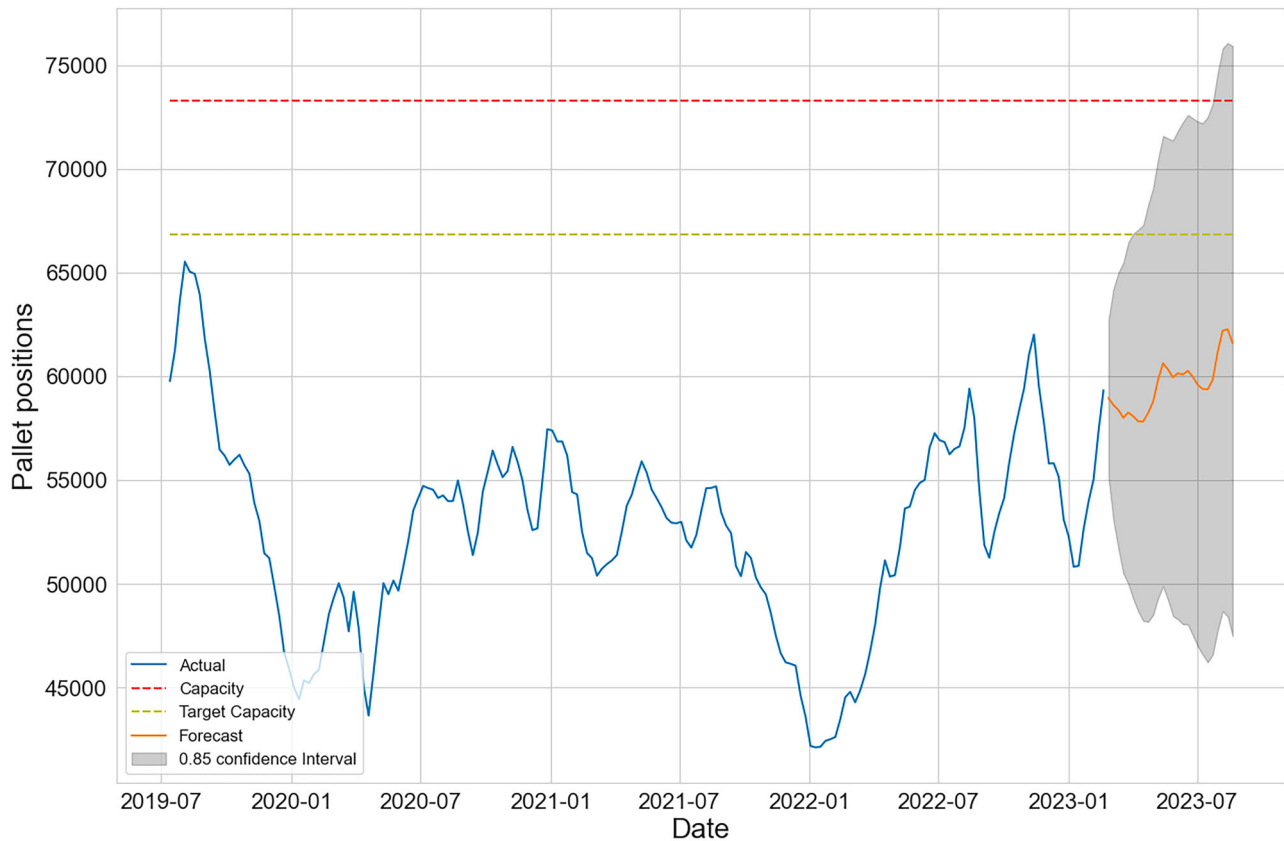


Figure 3. Forecast at site level performed by SARIMA $m(12) (1,0,1) (1,1,1)$. The figure contains the threshold for total and target capacities measured in pallet positions, historical time-series data, 26-week forecast, and credible interval.

The cooling chambers, in total, occupy enough space to allocate 13,632 pallet positions, with the target capacity at 13,216 pallet positions. Since one of the chambers was converted to a freezer on November 12, 2022, both the total and target capacity of a cooler decreased (Figure 5). The algorithm forecasts that on August 20, 2023, there should be customer demand for cooling chambers equivalent to 8008 pallet positions. Given the sigma of 3684 pallet positions, calculated using bootstrapped residuals, the probability of breaking the target capacity is 7.9%, and the probability of breaking the total capacity is 6.3%. Given the relatively modest risk of breaching capacity thresholds for cooling chambers, the data reinforces the need to further adapt existing facilities. Specifically, the conversion of one of the cooling chambers to freezing appears justified and strategically aligned with projected demands.

Even though the analysis indicates a need to expand the site, it's essential to recognise that the planning and construction process might extend over a year. This time-frame emphasises the urgency for immediate action. With the site hosting 385 individual customers, a one-size-fits-all approach is inappropriate since not all customers equally impact the total revenue. Considering the capacity constraints, it becomes crucial to further

apply the capacity planning framework to enable the segmentation of customers based on various criteria and facilitate decisive action. Decisions must be made regarding which customers should be prioritised with reserved capacity and which may need to be relocated to different sites, charged above the prevailing tariff, or potentially even terminated. This strategic approach will optimise utilisation and align customer management with both immediate needs and long-term projections.

4.2. Customer segmentation

After setting up the local database and conducting the data cleaning procedures, we were able to identify 385 individual customers who possessed stock at the site during the period covered by the dataset. This posed a significant challenge since generating 385 distinct customer forecasts would result in significant forecasting inaccuracies and render the outputs incomprehensible.

Segmenting was deemed crucial for generating useful forecasts for the organisation due to the presence of new customers annually and the overall large number of customers. Using the Customer Segmentation Matrix described in Section 3.3 with the data provided by Americold, we generated the output shown in Figure 6, where

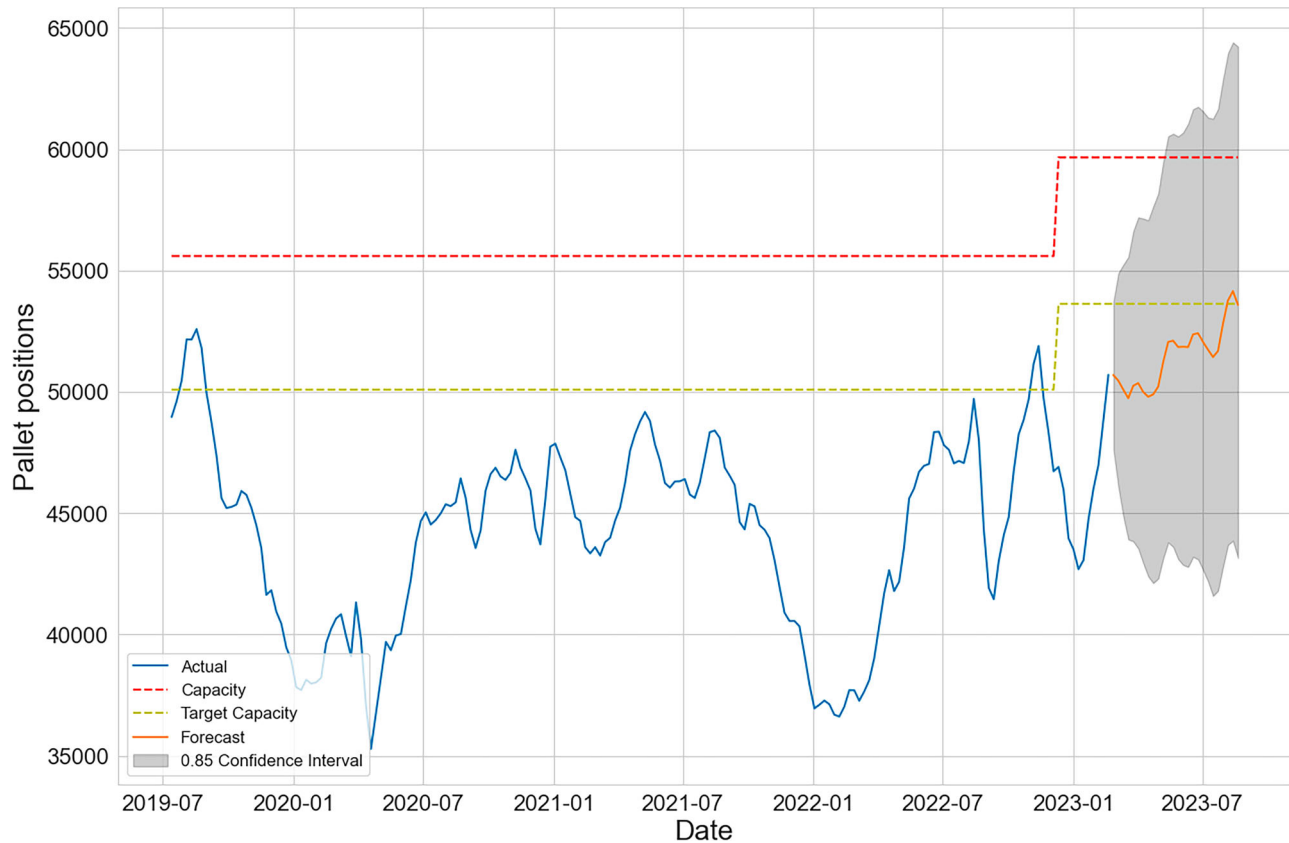


Figure 4. Forecast for freezer performed by Prophet. The figure contains the threshold for total and target capacities measured in pallet positions, historical time-series data, 26-week forecast, and credible interval.

each point represents all 385 customers in the data set. The Y-axis displays the average pallets for the analysis year, which was identified as 2022 because the current date was less than 6 months into 2023. In 2022, the largest customer had an average of 5521 pallets on hand. The X-axis shows the CV for each customer in the dataset, calculated across the entire data set.

To finalise the segmentation for each customer, three additional features were required for the proper scaling of the model. Firstly, we identified the storage temperature for each customer item, which allowed us to categorise all items into Freezer, Cooler, or Dry. Secondly, we identified customers within the multi-vendor consolidation program (MVC).

Multivendor Consolidation (MVC) is a strategic approach in supply chain management where shipments from multiple vendors are consolidated into a single, larger shipment (Glock and Kim 2014). It is particularly beneficial when a large retailer, such as Walmart, sources products from various suppliers in the same region. By employing MVC, companies like Americold assist low-volume customers in achieving cost efficiencies and improved service levels. This consolidation not only reduces transportation costs by transitioning

from multiple Less Than Truckload shipments to more cost-effective full truckloads but also ensures fewer delays, leading to consistent and reliable delivery times. Such a strategy is crucial for meeting stringent requirements, like Walmart's on-time in-full (OTIF) standards (Bower 2021). Through MVC, suppliers can also benefit from better transportation rates and services, fostering enhanced relationships and ensuring smoother operations in complex retail environments.

Regardless of their category, customers were given an MVC program flag along with the temperature stored. Lastly, we decided to treat key accounts individually due to their importance to the company (e.g. key customer 1). Americold believes it was important to track these customers individually due to their significance to the site. With these features, we were able to reduce the total number of unique customers from 385 to 12 actionable segmentation groups using this new customer segmentation model (Figure 7).

The conducted segmentation allows us to forecast capacity utilisation for each customer individually, which is a game-changer in dealing with capacity shortages. Performing the forecast for each customer segment individually uncovered some compelling trends and patterns.

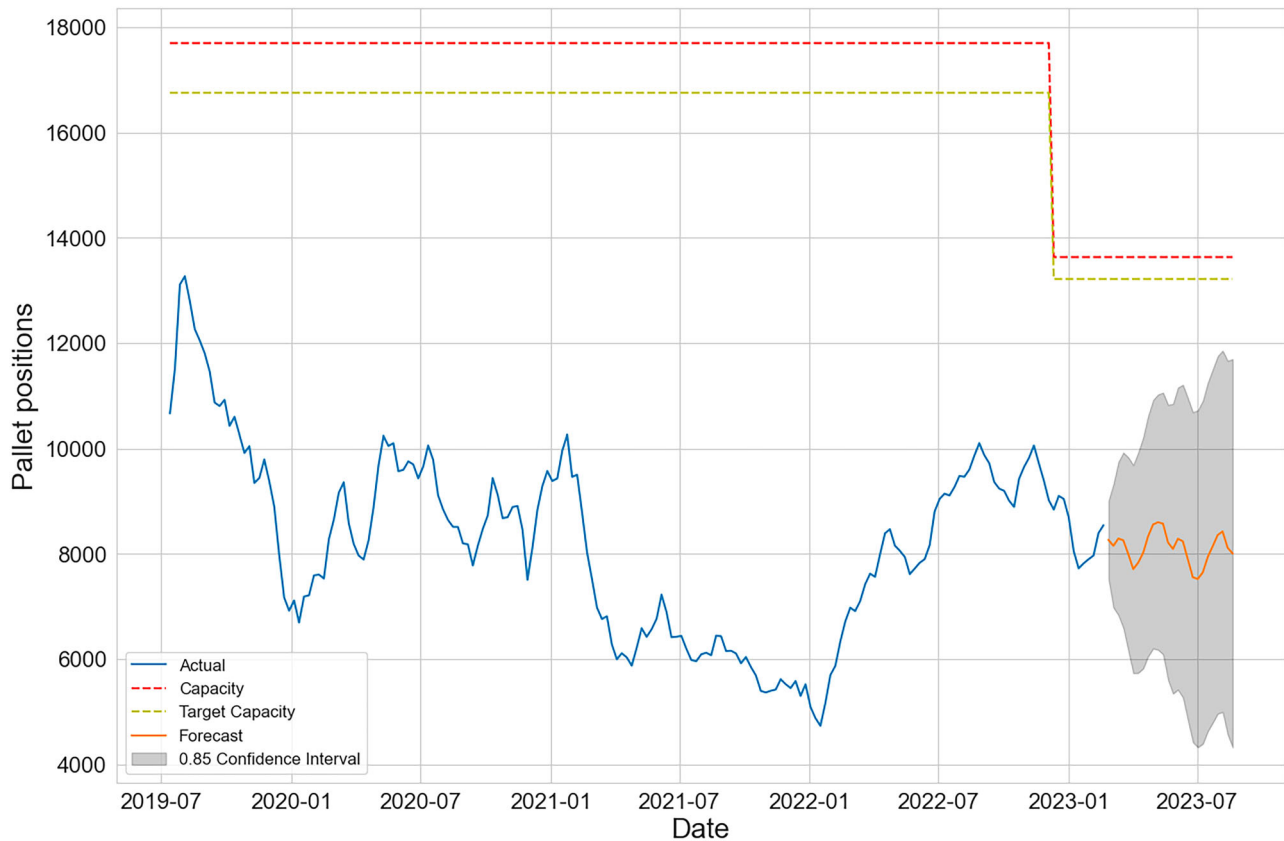


Figure 5. Forecast for cooler performed by SARIMA $m(52) (0,1,0)(1,0,0)$. The figure contains the threshold for totaland target capacities measured in pallet positions, historical time-series data, 26-week forecast, and credible interval.

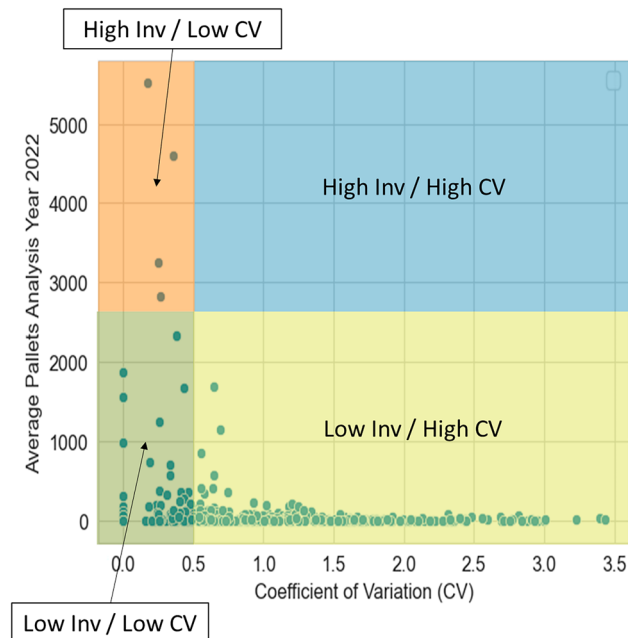


Figure 6. Scatterplot showing average pallets in 2022 (Y-axis) vs. the Coefficient of Variation (X-axis) for each customer within the data set overlaid with the customer segments created (Low Inv/ Low CV, High Inv / Low CV, Low Inv / High CV, High Inv / High CV).

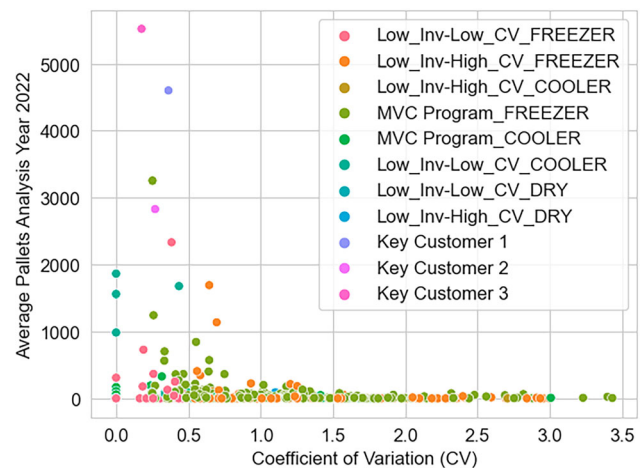


Figure 7. Final Customer Segmentation. Scatterplot showing average pallets in 2022 (Y-axis) vs. the Coefficient of Variation (X-axis) for each customer within the data set, with each customer colour-coded based on their customer segments.

Figures 8 and 9 provide us with a more detailed understanding of the cooler business, which can be used to develop effective business strategies. Upon analysing Figure 8, it is further evidence that the cooler business has



Figure 8. Time series showing the actual and a 26-week forecast for each customer segment within the cooler. The blue dotted line separates the actual data from the forecasted future.

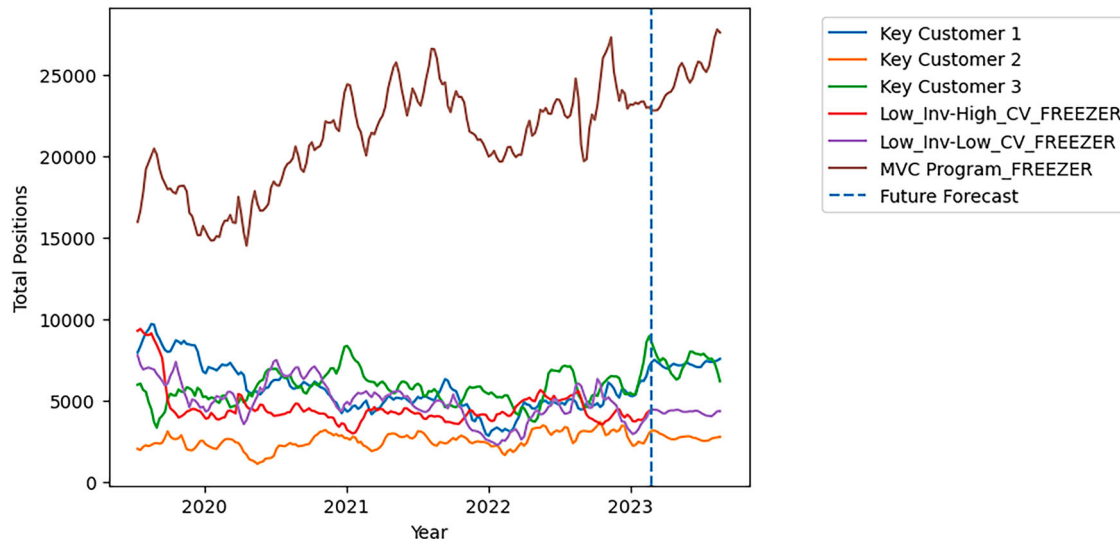


Figure 9. Time series showing the actual and 26-week forecast for each customer segment within the freezer. The blue dotted line separates the actual data from the forecasted future.

not shown growth over the last few years. The ‘Low_Inv-Low_CV_Cooler’ segment, in particular, has driven the flat to a negative growth trend that persists in the forecast. This lack of growth highlights the potential need for strategic interventions to improve the performance of this customer segment or find new customers that could help fill the inventory positions available.

In Figure 9, a significant finding was that the ‘MVC Program_FREEZER’ customer segment is projected to experience steady growth over the next 6 months. On the other hand, the remaining customer segments are mostly expected to remain flat or decline. These forecasted trends suggest that targeted interventions might

be needed to stimulate growth in the freezer customer segments that are expected to remain flat. Overall, this analysis reveals important insights that can be leveraged to develop effective business strategies.

Dry (ambient) customers account for the smallest portion of pallet positions. Americold does not prioritise ambient customers strategically, and these pallet positions were made available primarily because existing customers required a small amount of ambient space. As of 03/01/2023, there are no more available Dry (ambient) pallet positions, and there are no plans to create more in the future. Hence, don’t conduct any further analysis on this category.

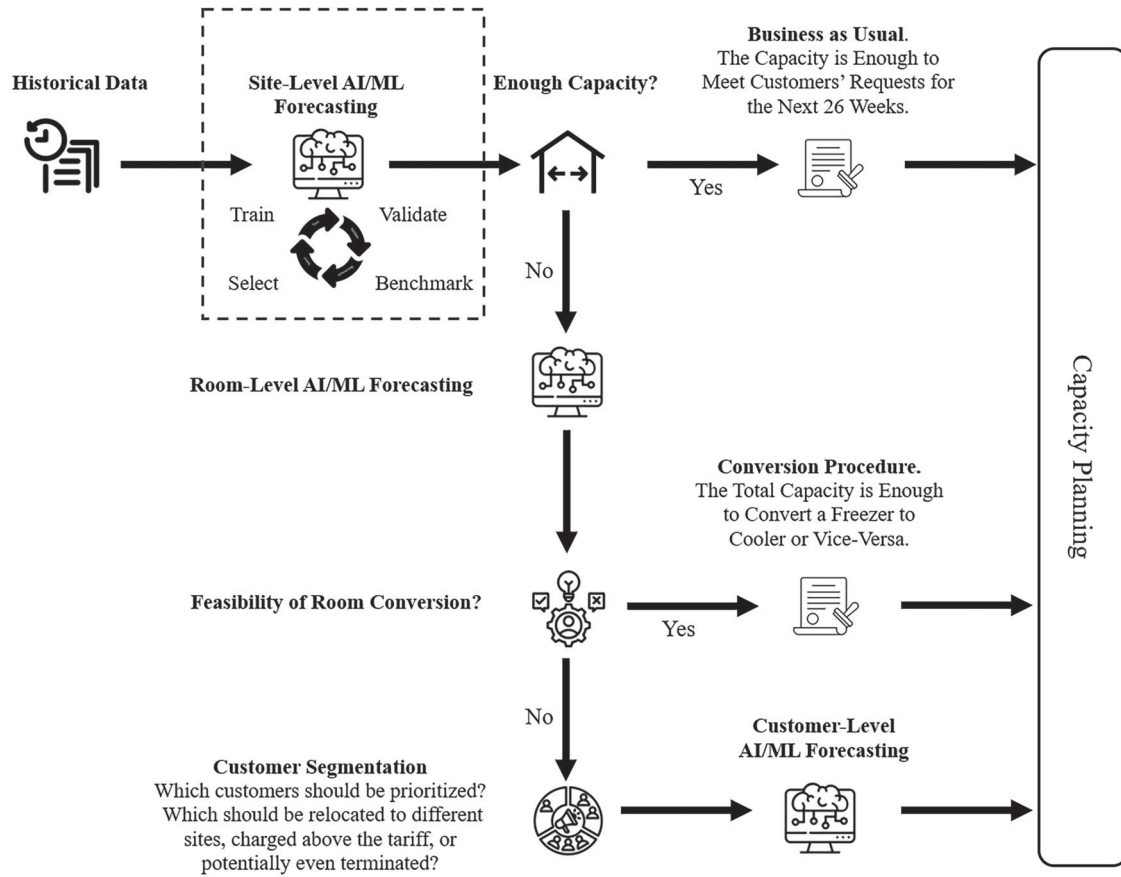


Figure 10. The proposed framework for Cold Chain Capacity Planning.

Table 1 summarises the accuracy of forecasting models for different customer segments. At first glance, the average difference of 1.7% in accuracy between the best and second-best models might seem negligible. However, such a difference can have profound implications in the context of supply chain management and inventory optimisation. Given the current scenario of ongoing capacity shortages, even minor inaccuracies in forecasting can lead to significant operational challenges. A seemingly small forecasting error can result in stockouts or over-stock situations, both of which can have detrimental effects on customer satisfaction and operational costs. In such a tight market, businesses cannot afford to have excess inventory occupying valuable storage space, nor can they risk running out of stock and losing sales opportunities. Furthermore, recent estimates from McKinsey underscore the financial implications of these discrepancies in forecasting accuracy. For fast-moving items, a mere 5% dip in forecasting accuracy can lead to a 2.5% surge in inventory costs, which ties up capital and increases the costs associated with storage, handling, and potential obsolescence. Moreover, the same dip can result in a 1.5% drop in revenue due to missed sales opportunities and decreased customer trust (Chui, Manyika, and Miremadi 2018).

5. Discussion

5.1. Framework for capacity planning

As previously mentioned, capacity planning within the cold chain domain is a challenging task due to the potential economic consequences stemming from inadequate cold chain capacity. The lack of capacity could result in substantial financial losses, legal complications, as well as adverse impacts on reputation and necessitate product recalls (Nerbovig 2017; Tsang et al. 2018). Additionally, distinct from the swift capacity adaptations observed in conventional supply chains, the task of changing capacity within the cold chain context is intricate and time-consuming. Moreover, the financial commitments associated with refrigeration infrastructure expansion are notably substantial. Such expansions are vital within cold chains as refrigeration facilities play a crucial role in preserving the quality and freshness of their commodities across supply networks (Min Wang and Zhao 2021).

Considering the distinctive attributes inherent in the cold chain, managers are faced with the necessity of formulating both short-term actionable planes and long-term strategic plans—particularly when the impending break of capacity is evident (Figure 10). This study

contributes a distinct vantage point primarily focussed on short-term actionable plans. Firstly, the study underscores that the strategy of uniformly reducing overall customer orders when approaching maximum capacity is inadvisable. This is due to the inherent variability among customers in terms of revenue generation and product classifications (e.g. freezer, cold storage, etc.). Secondly, an observation emerges that customer segmentation yields multifaceted advantages encompassing revenue management and capacity allocation. Notably, in the context of capacity management, the research reveals the absence of a universally applicable forecasting technique spanning all customer categories.

This framework serves as a guiding compass for identifying and distinguishing between customers who contribute significantly to the bottom line (in terms of revenue) and those whose contributions are relatively less substantial. This classification holds intrinsic value, as it empowers managers with insights to make informed decisions when it comes to optimising the allocation of precious capacity resources. In this manner, the capacity that might have been underutilised by less lucrative customers can be strategically repurposed to cater to the demands of the most lucrative customers, depicted in the top-left quadrant of Figure 6.

Furthermore, the framework's utility transcends mere capacity management, extending to a profound impact on pricing strategies. By identifying less lucrative customers, managers can implement pricing adjustments tailored to encourage these customers to migrate from a lower revenue contribution (i.e. the right quadrants depicted in Figure 6), to a more lucrative one, ultimately depicted in the top-left quadrant of Figure 6. This strategic pricing shift, aligned with the principles of the framework, not only addresses the immediate challenge of capacity shortages but also actively helps revenue optimisation. Hence, the comprehensive framework serves as an invaluable tool. It not only aids in optimising capacity allocation but also offers a blueprint to strategically realign pricing structures. By harmonising these crucial aspects, the framework serves as a multifaceted solution that navigates the complexities unique to the cold chain industry while maximising both operational efficiency and financial gains.

5.2. Managerial implications

Integrating AI/ML methodologies in cold chain management has opened new horizons for improving efficiency and planning. For instance, the use of the Customer Segmentation Matrix enables more accurate forecasting and planning for different customer segments, addressing the complexity of a diverse customer base. By segmenting customers, companies can tailor their approaches to

individual needs and trends, thereby reducing forecast inaccuracies. Americold's collaboration in this study provides a tangible case study demonstrating how advanced technology can be leveraged to manage a vast network of facilities.

The landscape of cold chain management demands not only accurate forecasting but also a more holistic and dynamic approach. The proposed framework extends beyond mere forecasting, offering a comprehensive approach to capacity planning that aligns with organisational needs and market demands. A dynamic planning process enables businesses to be more agile and responsive to fluctuations in market trends and unexpected disruptions, such as black swan events (Taleb 2005). This research underscores the inherent uncertainty and limitations of forecasting. Therefore, managers should adopt a cautious and flexible approach to planning, constantly evaluating and updating models to adapt to ever-changing market dynamics.

5.3. Limitations and future research

Recognizing the specific characteristics of Americold, the study focuses on a company that operates one of the world's largest networks of temperature-controlled warehouses. Americold's extensive infrastructure and operational scale make it an ideal subject for examining the application of AI/ML in cold chain logistics. However, it is important to acknowledge that findings from this single site of a single company may have limitations when generalising to other contexts. Nevertheless, the insights gained from Americold's case can provide valuable lessons for other companies in the cold chain industry.

Just like many research studies, our study has certain limitations and also suggests interesting directions for future research. Firstly, we've focussed our study on just one location within Americold Cold Chain. It would be valuable to examine how well our proposed framework works in different locations with diverse geographical settings. Secondly, we've divided customers into four groups based on their inventory and coefficient of variation (CV), as shown in Figure 6. Expanding the way we categorise customers could be worth exploring, for instance, by also considering the types of products they deal with. This might provide a more comprehensive understanding of how different factors influence customer behaviour and could offer more specific insights for both Americold Cold Chain and similar businesses. Future research should also consider the potential of cutting-edge technologies such as quantum computing to further enhance capacity planning and optimisation in cold chain logistics (Awan et al. 2022). Lastly, in future research, it could be beneficial to consider the problems

related to the planning of limited capacity in a broader theoretical landscape of the shortage economy (Ivanov and Dolgui 2022).

6. Conclusion

This study aimed to explore the application of AI/ML forecasting methodologies for capacity planning in cold chains, specifically within the context of Americold, the world's second-largest refrigerated LSP. Despite the advancements in AI/ML applications in supply chain management, there has been a noticeable gap in their use for optimising cold chain capacity planning. Previous studies have focussed mainly on logistical models and the adoption of technologies like blockchain and IoT, but they often overlook the unique complexities of cold chains. By developing and validating an AI/ML forecasting framework tailored for cold chains, our research addresses this gap and provides a more accurate and efficient approach to capacity planning.

Our findings demonstrate that AI/ML models, such as Facebook Prophet as well as traditional models like SARIMA, can significantly enhance demand forecasting accuracy. The proposed Customer Segmentation Matrix enabled more accurate forecasting and planning across various customer segments, addressing the issue of forecasting inaccuracies. These insights have practical implications for optimising resource utilisation and improving overall cold chain operations, thereby enhancing the integrity and quality of temperature-sensitive products.

Notes

1. <https://www.coldchainconference.org>
2. <https://ir.americold.com/financials/sec-filings/default.aspx>

Disclosure statement

No potential conflict of interest was reported by the author(s).

Data availability statement

The authors confirm that the data supporting the findings of this study are available and the subsample of anonymized data can be provided upon reasonable request.

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